SUPPLY CHAIN ANALYSIS REPORT

Field	Details
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Contents

1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through January 2018, focusing on identifying root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business $2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached $7.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders)	$7.5M
Profit at Risk (delayed orders)	$2.1M
Predictive Model Accuracy (RF)	74%

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2. Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at point of purchase.

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3. Analytical Objectives

- Understand the current state of delivery performance across all dimensions (region, mode, time, segment)
- Quantify the financial impact of delays on order profitability
- Identify the primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on-time delivery and profitability

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4. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

Metric	Value
Total Orders	172,765
Late Deliveries	94,523
Late Delivery %	54.71%
On-Time Delivery %	45.29%
Total Profit (profitable orders)	$7.5M
Profit at Risk (delayed orders)	$2.1M
90th Percentile Delay	3 days
Avg Order Profit	$22.03

• Late Delivery Rate at 54.71%

More than half of all orders arrive late. This is not an edge-case problem — it is the default experience for the majority of customers.

• $2.1M Profit at Risk

Orders that experienced delays collectively generated $2.1M in profit that is under constant pressure from further operational deterioration.

• 90th Percentile Delay = 3 Days

Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.

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5. Profitability Analysis

5.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

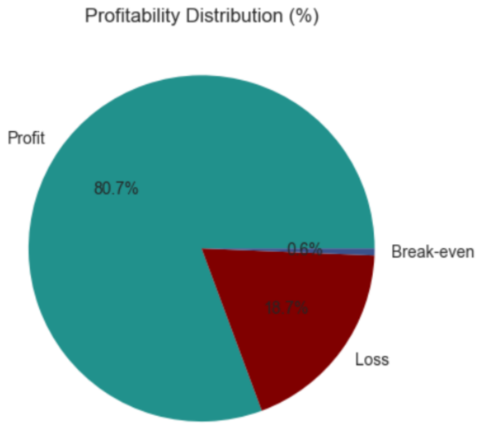

Fig i) Profitability Distribution

5.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late, the single largest cohort. Combined, orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

A critical observation: mean profit per order remains stable at approximately $21–$23 across all delay levels. The profitability problem is therefore driven by volume of delayed orders, not by individual order economics — making systemic throughput improvement the highest-leverage intervention.

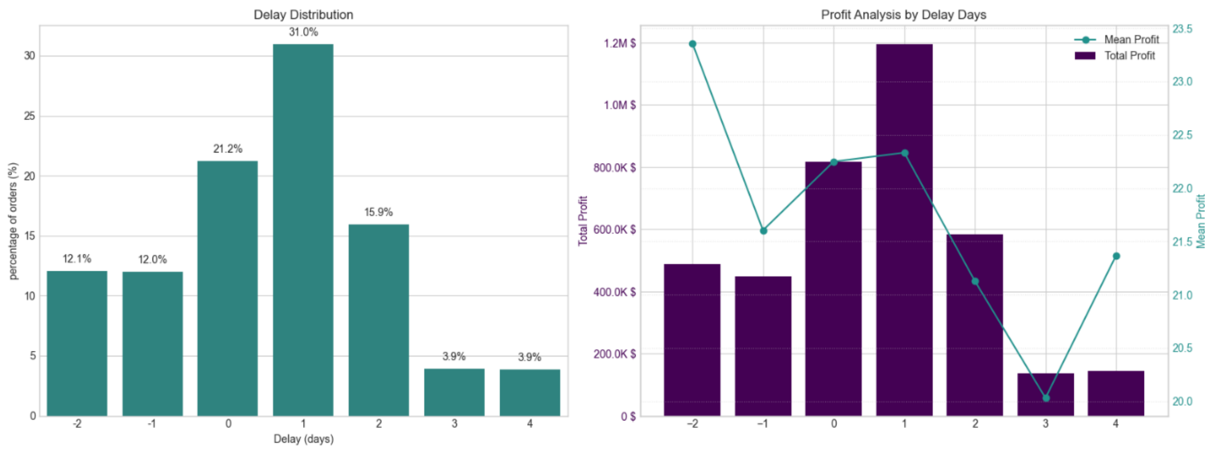

Fig ii) Delay Distribution & Profit vs. Delay Days

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6. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfillment process breaks down most severely.

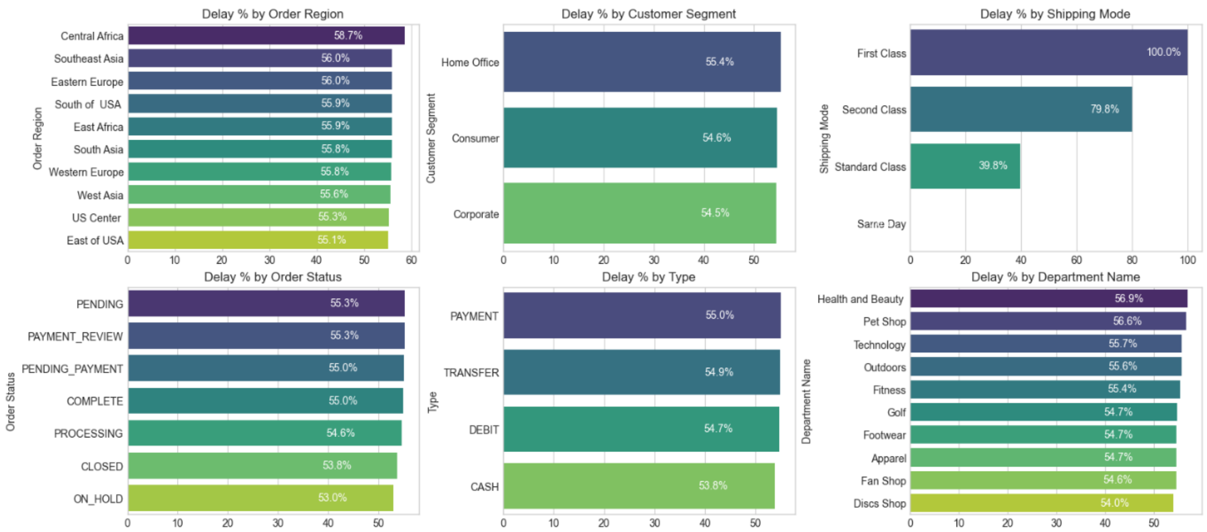

Figure iii) Delay % across six operational categories — Region, Segment, Shipping Mode, Order Status, Payment Type, Department

• Shipping Mode is the #1 Lever

First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. The mode assignment logic is the most impactful single variable to fix.

• Regional Spread is Narrow (55–59%)

Central Africa leads at 58.7% but all regions cluster between 55–59%. This rules out a localised logistics failure and confirms a company-wide systemic issue.

• Customer Segments Nearly Identical (54.5–55.4%)

No segment receives preferential service. All customers — Consumer, Corporate, Home Office — experience the same broken delivery promise.

• Health & Beauty (56.9%) and Pet Shop (56.6%) Lead by Department

These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.

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7. Root Cause Analysis

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region.

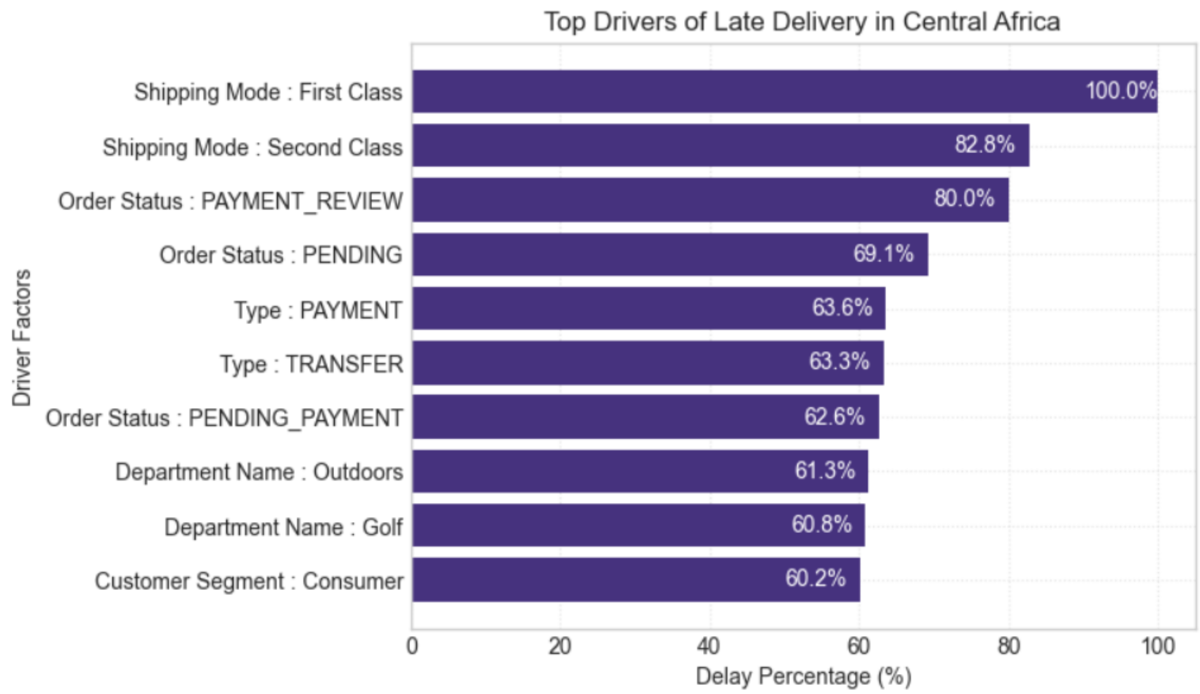

Fig iv) Top 10 drivers in Central Africa ranked by delay percentage

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8. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (53–57%), specific peaks highlight opportunities for targeted capacity planning.

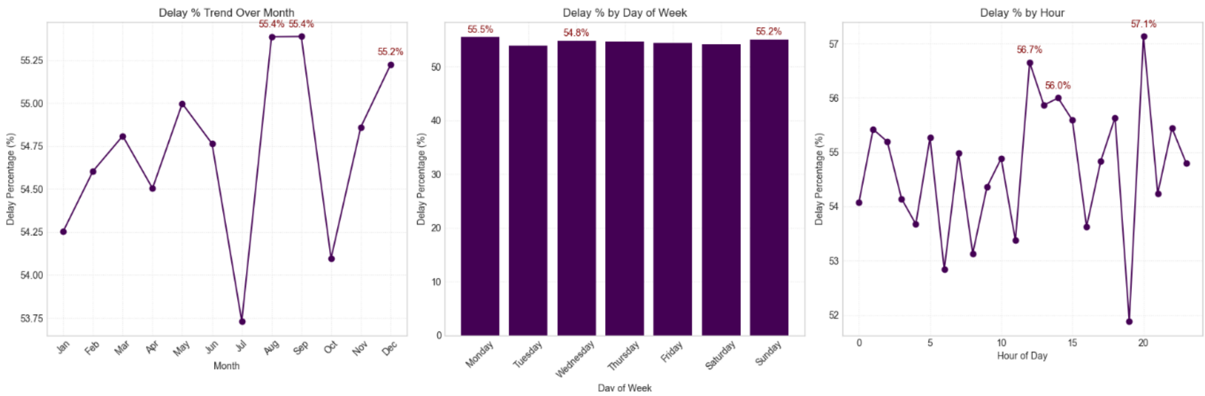

Figure v: Delay % by Month (left), Day of Week (centre), and Hour of Day (right) — red annotations mark the top 3 peaks in each dimension

• Peak Delay Months: August & September (55.4%) and December (55.2%)

August and September are the highest months at 55.4%, with December close behind. These reflect mid-year promotions and Q4 holiday surge overwhelming fulfillment capacity. July represents a notable low-point (~53.75%), confirming seasonal variation is plannable.

• Day-of-Week Variance is Minimal

Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day-of-week is not a meaningful lever for operational intervention.

• Intra-Day Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)

Late-evening orders (hour 21) have the highest delay rate at 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.

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9. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variables, stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On-time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553

• 74% Overall Accuracy

The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late-delivery alert system.

• Higher Precision on Late Orders (0.78)

When the model predicts an order will be late, it is correct 78% of the time — sufficient to trigger targeted interventions without generating excessive false alarms.

• Recall for Late Orders at 0.75

The model captures 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential.

• Class 0 Harder to Predict (Precision 0.68)

On-time deliveries share many characteristics with late ones given the systemic nature of delays, making them genuinely difficult to distinguish.

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10. Strategic Recommendations

Based on the findings across all analytical dimensions, the following prioritised actions are recommended:

Priority	Recommendation	Description
CRITICAL	Audit First & Second Class Shipping	First Class has a 100% delay rate and Second Class 79.8%, contradicting their promise. Conduct immediate SLA audit; consider suspending or repricing.
HIGH	Deploy Predictive Alert System	Use Random Forest model (74% accuracy, 78% precision) to flag high-risk orders and trigger proactive actions.
HIGH	Resolve Payment Bottlenecks	PAYMENT_REVIEW (55.3%) and PENDING_PAYMENT (55.0%) orders show high delays. Implement escalation for delays >2 hours.
MEDIUM	Seasonal Capacity Planning	August, October, December show peak delays (~55%). Plan capacity, partnerships, and adjust delivery timelines.
MEDIUM	Default to Standard Class	Standard Class has lower delay rate (39.8%). Optimize shipping mode allocation.
MEDIUM	Investigate African Departments	Outdoors (61.3%) and Golf (60.8%) show high delays. Audit inventory and logistics operations.
LOW	Reduce Loss-Making Orders	18.7% of orders are loss-making. Review pricing and discounts, especially for delayed orders.
LOW	Retrain Predictive Model	Retrain quarterly; include additional features (weather, carrier data). Target >80% recall.

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11. Conclusion

This analysis has surfaced a clear and urgent picture: a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, costing $2.1M in at-risk profit and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping mode misconfiguration (First Class at 100% late, Second Class at 79.8%) is the dominant operational failure. Payment processing friction creates a secondary bottleneck. Seasonal volume spikes are not adequately planned for. And geographic disparities, while present, are secondary to the systemic company-wide pattern.

A Random Forest predictive model trained on readily available order features already achieves 74% accuracy in identifying at-risk orders. This is a deployable tool, not a future aspiration.

Priority Area	Current State	Target State
Late Delivery Rate	54.71%	< 30% within 12 months
First Class On-Time Rate	0%	> 80%
Second Class On-Time Rate	20.2%	> 60%
Predictive Model Accuracy	74%	> 82% (enhanced model)
Loss-Making Orders	18.7%	< 12%
Profit at Risk	$2.1M	Reduce by 40%
